# 4AL3 Project Proposal

October 9, 2026

Group 10

Ali Zia (ziam8), Andrew Habib (habiba21), Gurmanjot Minhas (minhag7)

Consulted TA: Yasmin (End of Tutorial 3, 12:30pm section)

**1. Team Members:**

Ali Zia, Andrew Habib, Gurmanjot Minhas

**2. Task Title and Overview:**

Our task is titled “Expected Goals (xG) Prediction in Soccer”. The goal is to predict the probability of a shot in a soccer match resulting in a goal using surrounding context as features (i.e. location on the pitch, body part, technique used, position of nearby defenders and teammates) to predict the probability that it results in a goal. The significance of this task lies in its relevance to modern soccer analytics. The xG metric is one of the most used metrics in soccer by analysts, clubs, and commentators to evaluate the quality of a scoring chance independently of the outcome of whether it went in. While both a tap-in and a 30-yard free kick may result in a "no goal" outcome, they represent different levels of play and difficulty that xG can help differentiate. Overall, this task is foundational to practical applications in soccer analytics as it underlies player performance evaluations which further influence team strategies. The biggest challenge in this task relates to determining how much shot features contribute to danger levels and how they impact the chances of scoring a goal. For instance, two shots can be taken from the same location using the same technique, but one can be a clear goal-scoring chance while the other can be heavily contested based on the location of defenders. Overall, building a model that generalizes well will be challenging due to the varying feature interactions plus the fact of goals being a rare outcome relative to shots taken in the sport as a whole.

**3. Task Definition:**

The data consists of structured event data that describes individual shots taken during matches, including both quantitative features (i.e. shot location coordinates, defender positioning) and qualitative features (i.e. body part, shot technique). This task is an example of binary classification with two classes: goal or no goal. The underlying prediction is probabilistic since the output is best predicted as a likelihood of a goal rather than a concrete label. We will also be using a single-label approach where each shot is assigned the label of goal or no goal.

**4. Problem and Impact:**

The core problem revolves around how the raw measure of shot outcomes is a poor measure of a team’s or player’s finishing quality because of randomness and small margins. For instance, a crossbar hit from an open chance versus a weak shot from a tight angle that deflects off a defender and goes in, doesn’t accurately reflect the actual quality of chances that were created. Our approach is to model the probability a shot results in a goal based on the qualitative and quantitative context surrounding the

event so that the chance quality can be evaluated independent of the shot's outcome. xG is deeply embedded in soccer culture, which makes it impactful in the real world. Scouts may use a striker's cumulative xG versus their actual goals scored to judge whether they are over- or under-performing their chances, directly informing their transfer decisions worth millions of dollars. On the other hand, xG also benefits fans and commentary by giving neutral language to describe how a match played out, independent of the final score. The problem is challenging primarily because shot quality depends on many contextual factors. For instance, while distance and angle may be important, they can be greatly affected by the body part or technique used and the defender's positioning. Additionally, translating raw data of player coordinates into a meaningful conclusion requires a lot of feature engineering. Lastly, the fact that goals are a rare event relative to shots taken adds a layer of complexity to discern chance quality from a dataset that is naturally skewed toward a "no-goal" outcome.

**5. Data Source & Plan for Data Collection:**

**Data Source:** We will use StatsBomb Open Data, an open-source dataset with documentation available in a public GitHub repository that contains football data from various competitions (1 JSON file per match). No web scraping or API access/rate limiting is required. For terms of service, we will adhere to the StatsBomb Public Data User Agreement, and we will credit StatsBomb as the data source.

**Data Labelling:** The data has shot outcomes (target label) for each shot event. No further labelling work is required. We only need to convert the target label to Goal (1) or No Goal (0) instead of the current multi-label classification and parse the JSON into a clear tabular format.

**Metadata:** Since the only relevant folder is events, the following are some metadata from those JSON files: Match ID (Denoted in the JSON file name), Event ID (identifier for each event in the match), Period (Part of the match, i.e., first half), Timestamp/Minute/Second (time the match event takes place), Event Type (In our case, we are looking for "shot"), team, player, and more.

**How data was collected, annotated, and preprocessed:** The GitHub page states that the JSON data was exported from the StatsBomb Data API. According to the official StatsBomb/Hudl page, they collect data using cameras and Computer Vision to capture live footage tracking each event. They make JSON, XML, and CSV data available through their APIs.

**Subset or Complete Dataset and Features:** Only data relevant to "shot" events listed in the JSON files of the events folder will be used in its entirety. The events folder contains 3235 JSON files, each depicting a distinct match, with each match having around 20-40 shots on average, which is highly

manageable to use in its entirety without compute becoming an issue. All other unrelated JSON content is discarded because it does not have a clear relationship to goals. (See 6 and 7 for dataset size and features/target label). Within the relevant "shot" data, metadata such as player details (name, position, number) can be discarded, while information such as location, technique, deflection, body part, etc. directly impacts whether a goal can be scored.

**6. Expected Size of Dataset:**

**Expected size of data:** 108,000 shots across 3235 matches. We will use a 70/15/15 train-validate-test split, which provides 75600 shots, 16200 shots, and 16200 shots, respectively.

**Data Point Examples:** *Note: Examples do not contain full data point information since it is highly comprehensive and requires parsing. Below is a summary of sample features.

- Shot from Match 19781 (Outcome/Label: Goal) → Location: [111.0, 43.0], Type: Open Play, Body Part: Left foot, Technique: Normal, Sample locations of other players: { location: [117.0, 45.0] (teammate = false), location: [118.0, 61.0] (teammate = true) }
- Shot from Match 9609 (Outcome: Saved = No Goal) → Location: [107.5, 45.6], Type: Open Play, Body Part: Head, Technique: Normal, Sample locations of other players: { location: [96.5, 13.7] (teammate = true), location: [92.1, 33.1] (teammate = false) }
- Shot from Match 19745 (Outcome/Label: Blocked = No Goal) → Location: [109.0, 45.0], Type: Open Play, Body Part: Head, Technique: Normal, Sample locations of other players: { location: [110.0, 46.0] (teammate = false) }

**7. Proposed Solution:**

To solve this problem, we plan to use logistic regression (LR) to determine the probability of a goal based on its conditions (see Features and Target Labels), and further classify the chance by its danger level so it is more readable for unfamiliar observers. Other viable models we may consider include support vector machines (SVM) and decision trees (DT) [1]. For SVM, our idea is to calculate the distance of some desired point from the hyperplane and use a function to convert this into a probability or danger level. For DT, each parameter would be decomposed into ranges or buckets. The training data would then assign each leaf node a total shot count and goal count, from which a corresponding probability could be calculated (# goals / # shots) for similar circumstance shots. Existing industry solutions such as Opta and StatsBomb use extreme gradient boosting models (XGBoost) [2] [3] because they independently assess nonlinear feature combinations (i.e. preferred foot + volley vs weak foot + volley) with decision trees and repeatedly refine them: distinctions that are harder for LR to map

to a function. However, decision trees are prone to overfitting and inherently struggle with continuous features (i.e. shot distance) [4], so LR is our best option until XGBoost is covered in the course.

One way we will evaluate the model is by assessing the calibration of model predictions. If a model assigns a set of chances an xG of 0.3, approximately 30% should result in goals. The closer the predicted probabilities are to the actual goal frequencies, the better calibrated the model is. Another way we will evaluate it is by calculating the log loss score. This score penalizes wrong probability predictions, with larger penalties as the predictions get worse, so it makes sense to minimize this when tuning any hyperparameters during the validation phase.

The dataset is very rich with spatial features such as relevant teammate/opponent location coordinates and shot start/end locations. It details shot features like the technique, body part, action leading up to the shot (i.e. aerial duel, dribble, first-time shot), and whether there was a deflection or open goal. Finally, it also labels whether the shot resulted in a goal or not. Individually, these features are useful but will need to be engineered to account for more relevant factors like shot angle and defensive pressure [1]. The dataset contains information about all passes and touches, so these events can be processed to determine a player's preferred footedness and how it impacts their shot outcome as well. Once a goal probability is determined, each shot will be classified into one of the following danger levels: Low (0.00-0.10), Medium (0.10-0.30), High (0.30-0.50), Very High (0.50-0.75), Extreme (0.75-1.00).

For the current LR approach, Numpy and Scikit Learn will be sufficient, but the XGBoost and Pytorch libraries may also be needed if we pivot to more advanced strategies. Also, since the dataset exists as JSON files in a GitHub repository, the Requests and JSON libraries will be needed for the pre-processing stage.

**8. TA Discussion (TA: Yasmin):**

We talked to the TA about the feasibility among 2 ideas: an email classifier and soccer goal prediction. She verified each problem was nontrivial and achievable within the scope of the course. To help us narrow down the ideas, we were advised to verify the quality of each dataset and make sure there was enough data to train a model. Based on this advice, we ruled out the email classifier idea as it used a synthetic dataset, influencing our plans to choose the soccer goal predictor. As this dataset is contained in JSON files, we were also advised to pre-process relevant features into a tabular format which will influence our strategy for early stages of data collection and model training.

## 9. References:


[1] J. H. Hewitt and O. Karakuş, "A machine learning approach for player and position adjusted expected goals in football (soccer)," *Franklin Open*, vol. 4, Art. no. 100034, 2023, doi: 10.1016/j.fraope.2023.100034.

[2] J. Whitmore, "What is expected goals (xG)?," *Opta Analyst*, Aug. 8, 2023. [Online]. Available: What Is Expected Goals (xG)?.

[3] Hudl StatsBomb, "StatsBomb global football data glossary," *Hudl Support*. [Online]. Available: StatsBomb Global Football Data Glossary.

[4] D. Okello, "XGBoost vs decision trees: A comparative overview," *DEV Community*, Nov. 21, 2024. [Online]. Available: https://dev.to/okellodaniel/xgboost-vs-decision-trees-1gg0.